PROTODUINO · DESIGN DOCUMENT

A streaming ELF loader for AVR

Functional and technical design of a compile-time configurable ELF32 loader: bootloader core, flash service and loadable processes. With measurements from a working prototype and fixes for current protoduino caveats.

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84 B

of loader state on an ATmega328P; no heap, no seeking

1,676 B

serial ELF bootloader for the ATmega328P, as measured

5.6 %

of example 12's ELF file is all the loader has to receive

12 of 12

simulated scenarios on ATmega328P and ATmega2560 behaved as designed

What is real in this document

- A prototype of the loader core and of a serial bootloader built on it was written and run for this document. Every figure marked MEASURED comes from it: avr-gcc 7.3.0, and simavr at the commit that protosim pins.
- Nothing has run on hardware. Power loss during an update was not simulated.
- The module layer (6.8), relocation (4.7) and the ATtiny path (6.6) are designs. The ATtiny build compiles; it has not executed anywhere.
- No protoduino or protosim source file was changed. The fixes in section 7 and section 8 are proposals. The prototype sits beside this document in `docs/elfloader/prototype/`, outside `src/`.

MEASURED built and run for this document READ established by reading source or tool output PROPOSED designed, not built BUILT compiles, not executed

Contents

1 Summary
2 Goals, non-goals and target parts
3 What earlier loaders got wrong
4 The ideas the design rests on
5 Functional specification
6 Technical design
7 Fixes for current protoduino caveats
8 What protosim needs
9 Verification
10 Roadmap, risks and open questions
A ELF fields the loader reads
B Reproducing the measurements · Sources

1 Summary

The loader takes the file the linker writes, the `.elf`, and treats it as a stream. The first 52 bytes say what the file is. The next 32 bytes per program header say which bytes belong where. Everything after that is delivered to flash in the order it arrives. The loader never seeks backwards and never allocates, and all of its state fits in one 84-byte structure. When the last loadable byte has been delivered it is done; the symbol tables and debug information that make up most of the file never have to be sent.

That one core is meant to be built three ways, chosen at compile time:

- **As a bootloader.** Core, a polled serial port and XMODEM-CRC in the boot section. Any terminal program can send the unmodified `.elf`. An image record written last decides whether the application may start.
- **As a flash service.** The same boot section exports a page-write entry, so the application can program flash although the SPM instruction only works from the boot section.
- **As a runtime loader.** Inside the kernel, the core installs modules into flash slots while the system runs. A module is a protoduino process type that lives in flash; starting it is `process_new()` plus `process_start()`.

The third use is what makes this fit protoduino in particular. Protothread processes keep their state in a struct the kernel hands them, so a module needs no static RAM, and the hardest part of dynamic loading on an AVR, relocating data addresses, does not arise.

Result	Value	Status
Loader state, ATmega328P build	84 bytes, of which 52 are header scratch	MEASURED
Serial ELF bootloader, ATmega328P	1,676 B flash, 349 B RAM; 1,944 B with every option on. Fits the 2 KB boot section, not the 1 KB one.	MEASURED
Serial ELF bootloader, ATmega2560	2,168 B flash, 491 B RAM; 2,488 B with every option. Needs the 4 KB boot section today.	MEASURED
ATtiny85 build with bit-banged serial	1,736 B flash, 285 B RAM	BUILT
Flash written by the loader against simavr's own ELF reader	Identical on both parts, including a 106 KB image that reaches far flash	MEASURED
A file for the wrong core family, offered to a device with a working application	Refused with a reason code; nothing erased; the application starts again	MEASURED
Application programs a flash page through the loader	Works in simulation; the simulator does not enforce the boot-section rule	MEASURED
A module that declares static RAM	Fails at link time with a clear message	MEASURED

The size is the weak point, and it should be stated plainly. Optiboot fits in under 512 bytes; this prototype is three times that, in plain C with an old compiler, and it does not fit a 1 KB boot section. Section 6.10 shows where the bytes go and what could shrink them. Reaching 1 KB is a goal, not a result.

2 Goals, non-goals and target parts

Goals

- **Accept the linker's own output.** No hex file, no binary image and no packaging step for firmware.
- **One core, cut at compile time.** From a bootloader that fits a boot section to a loader inside the kernel,

Non-goals

- **Linking on the device.** Relocatable `.o` files, symbol tables and name lookups stay on the host.
- **Authenticated updates in the first version.** Integrity is checked; origin is not. A place for a

with features that cost nothing when switched off.

- **Refuse before erasing.** The wrong file must be rejected while the installed image is still intact.
- **Always recoverable.** On parts with a hardware boot section, no failed update may leave a device that cannot be reached over the same serial line.
- **Modules are processes.** A loaded module is an ordinary protoduino process.

signature is reserved (6.8).

- **C++ runtime features in modules.** No exceptions, RTTI or static constructors.
- **Parts without room for a page buffer.** The ATtiny4/5/9/10 have 32 bytes of RAM.
- **UPDI-era parts in this document.** tinyAVR 0/1/2 and AVR Dx program flash through NVMCTRL. They need another flash back-end, which the layering allows but which is not designed here.

Part	Flash	RAM	Page	Boot section	Role in this design	Status
ATmega328P	32 KB	2 KB	128 B	yes, 0.5 to 4 KB	Recovery bootloader, service, modules	MEASURED
ATmega2560	256 KB	8 KB	256 B	yes, 1 to 8 KB	The same, with 24-bit addresses	MEASURED
ATmega1284P	128 KB	16 KB	256 B	yes	The same	BUILT
ATtiny85	8 KB	512 B	64 B	no	Updater behind a patched reset vector	BUILT
ATtiny25 / 45	2 / 4 KB	128 / 256 B	32 / 64 B	no	Too little RAM for a 128-byte XMODEM block; needs a small-frame transport	PROPOSED
ATtiny4/5/9/10	≤ 1 KB	32 B	–	no	Out of scope	–

Flash end, RAM range, page size and core family were read from the avr-libc device headers. Boot-section sizes are from the datasheets and were not re-checked here. The request named an “ATtiny8”; no such part exists, so this document takes the ATtiny25/45/85 family as the small target, with the ATtiny85 as reference.

3 What earlier loaders got wrong

Five existing designs were examined, four of them in source. Each got something right that this design keeps, and each left something that it changes.

Loader	What it does, and where it falls short	What this design does instead
Contiki ELF loader 2005 to 2017 READ	Loads relocatable `.o` files and links them on the device. It makes 14 seek-and-read calls, recognises ten sections by comparing name strings, and resolves every import with `strcmp` against a table of names compiled into the kernel. So the whole file must already sit in a seekable store, and the kernel carries its symbol names in flash. It checks the first seven identification bytes and never the machine type. Contiki later added a pre-linked format (CLE), and Contiki-NG deleted the loader in 2017, 4,203 lines in one commit.	No linking on the device. The file is consumed as it arrives. Machine and core family are checked from the header. What Contiki got right is kept: a module is a set of processes, and architecture code is separate from the core.

Loader	What it does, and where it falls short	What this design does instead
Optiboot with avrdude	Under 512 bytes, which is the bar for size. But the device receives anonymous pages: it has no idea what the image is, keeps no record of whether it is complete, and starts whatever is there after a timeout. The host has to turn the ELF into a hex file and speak a programmer protocol. Version 8 added `do_spm()` so an application can write flash.	The device reads the ELF header itself and keeps an image record. The `do_spm()` idea is kept and widened into a service entry (6.7).
urboot, micronucleus parts without a boot section	They patch the application's reset vector so that reset reaches the loader, and rely on erased flash running through to it. The known hazard, in urboot's own words: if the first page is erased but not rewritten, the part is bricked until it meets an ISP programmer.	Erase from the top down with page 0 last, and give the loader a landing pad. There is then no instant at which a reset fails to reach the loader (6.6). PROPOSED
simple-avr-bootloader m3y54m; a tutorial by its own description READ	Sends a raw binary with a 16-bit length. Frames carry an 8-bit additive checksum and no sequence number, and the uploader retries on timeout, so a lost acknowledgement makes the same chunk land twice. The update flag sits 8 bytes below the top of RAM, where the stack starts. The loader runs its serial port from interrupts and has no check that the application is whole.	XMODEM-CRC: 16-bit CRC and block numbers, so a repeat is acknowledged and dropped. A polled serial port with no vectors. An image record. Its good idea is kept: the application asks for an update, then resets.
Host ELF libraries libelf, ELFIO and others	They assume random access or the whole file in memory, and they allocate. None is built to be fed.	Used only as oracles in tests.
Zephyr LLEXT a current reference	Still links on the device, but can replace symbol-name strings with hashes to save space. That is the right direction, and still far too much code for an 8-bit part.	One step further: no symbol table on the device at all. A module is linked on the host against one firmware build and carries that build's identity (6.8).
protoduino `docs/loader.md` READ	Describes modules that are stored in RAM and run from there, which an AVR cannot do: code executes only from flash. It compiles with `-fpic`, which avr-gcc reports as unsupported. It refers to functions that are not in `src/`. Its file header has no magic number, version or checksum.	Superseded by this document.

An earlier note of mine questioned the order in which simple-avr-bootloader fills, erases and writes a page. That was wrong: the datasheet allows the page buffer to be filled either before or after the erase.

4 The ideas the design rests on

4.1 The ELF file is the wire format

An executable from avr-gcc is laid out in the order a loader needs it. The ELF header comes first, the program headers follow at offset 52, the loadable bytes come next, and everything a loader has no use for comes last. A reader that only moves forward can therefore do the whole job, and can stop early.



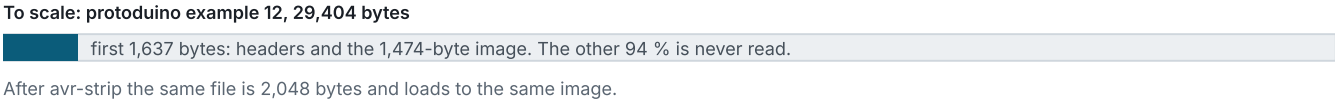


Figure 1. Everything the loader needs is at the front of the file. MEASURED

File	File size	Loader finished within	Result against `avr-objcopy`
Test program, ATmega328P, with EEPROM data, fuses and signature	12,860 B	660 B	Flash and EEPROM identical
protoduino example 12 (UART echo), ATmega328P	29,404 B	1,637 B	Identical, also when fed one byte at a time
106 KB test image, ATmega2560	117,616 B	106,986 B	Identical

The consequence for the user is that no tool has to be installed to flash a board. With XMODEM as the transport (6.4), a terminal program that every platform already has can send the file the build produced.

4.2 One core, three cuts

The core is a state machine with no input or output of its own. Bytes are pushed in from whatever carries them; bytes that belong in a memory are handed to a sink. Transport, sink and policy are separate pieces, and a build takes only the ones it needs.

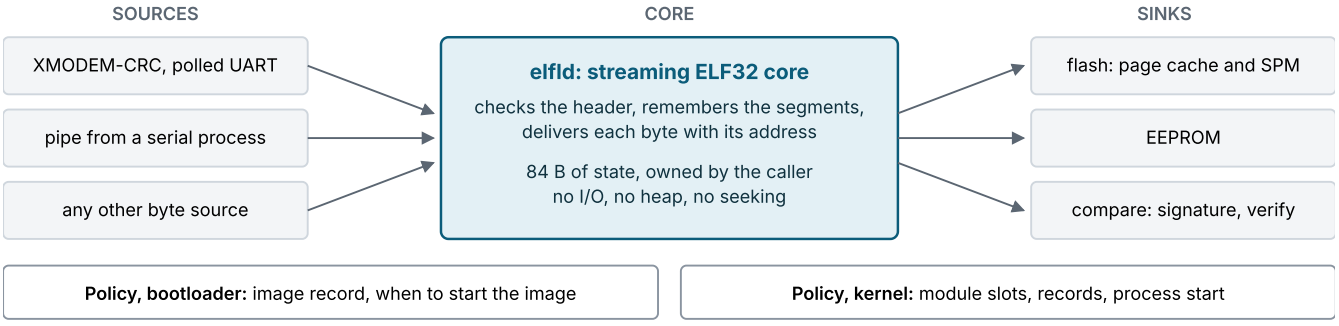


Figure 2. The pieces. The top row of each column is what the prototype contains.

Cut	Lives in	Contains
A. Bootloader	Boot section, or the top of flash on parts without one	Core, XMODEM receiver, polled serial, flash back-end, image record
B. Bootloader with service	Boot section, plus a 4-byte stub at the end of flash	Cut A, and a page-write entry the application may call
C. Runtime loader	Application section, as a protoduino process	Core and module policy; flash writes go through the service of cut B

4.3 Address spaces select the sink

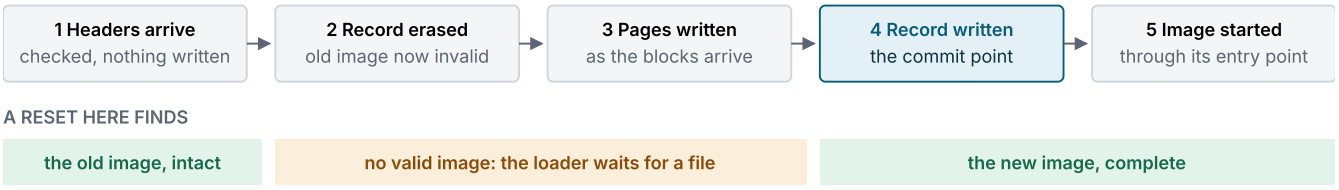
avr-gcc gives every memory of the chip its own range in the ELF address space, so the top byte of a segment's load address says where its bytes belong. The loader routes on that byte. It uses the load address (`p_paddr`), not the run address: the initial values of `.data` are stored in flash right after the code, and the program's own start-up copies them to RAM.

Load address	Memory	What the loader does	Option
`0x00....` upward	Flash	Writes, inside the permitted window only	always
`0x80....`	RAM	Nothing to write. `.bss` has no file bytes. The declared extent can be recorded.	`ELFLD_FEAT_RAM_EXTENT`
`0x81....`	EEPROM	Writes the EEPROM image	`ELFLD_FEAT_EEPROM`
`0x82....`, `0x83....`	Fuses, lock bits	Ignores; software cannot write them	–
`0x84....`	Signature	Compares with the chip's own signature bytes	`ELFLD_FEAT_SIGNATURE`

Two checks come free with the format. The header's flags name the core family the file was built for (`avr:5` for an ATmega328P, `avr:6` for an ATmega2560, `avr:25` for an ATtiny85), and avr-gcc defines the same number as `__AVR_ARCH__` when the loader is compiled. One comparison rejects a file for the wrong family from its first 52 bytes. The signature check is finer, down to the exact device, but it comes late, because avr-gcc places the signature after the flash data (see FR-9).

4.4 Refuse before erasing, commit last

An update is a transaction with one commit point. Nothing in flash changes until every program header has been accepted. The installed image is declared invalid before its first byte changes, and becomes valid again only when a small record is written after the last byte. A reset at any moment therefore finds either the old image, or no valid image and a waiting loader, or the new image.



A refusal in step 1 leaves the device exactly as it was. The first prototype got this wrong; test T7 found it.

Figure 3. The update as a transaction. Steps and refusals were exercised in simulation; a power cut was not. MEASURED

4.5 The boot section as a small ROM

On a megaAVR the SPM instruction works only from the boot section, so an application cannot program flash by itself, and a runtime loader in the application cannot exist without help. The boot section therefore exports a service: an entry at a fixed address at the very end of flash, with two magic bytes beside it, that programs one page from the caller's buffer and returns. The design goes one step past Optiboot's `do_spm()`: because the core keeps its state in a structure the caller owns, the same entry table can also export the core itself, and the kernel's module loader then costs policy code only (6.7).

4.6 A module is a process type

protoduino already has the right descriptor. `process_type_t` is a small record in flash holding a name, a thread function, an instance size and a priority, and `process_new(type, storage)` turns it into a running process in memory the caller supplies. A loadable module is exactly that record plus its code, linked into a flash slot. Loading a module means writing its bytes and calling the two functions every built-in process already uses.

Protothreads are stackless, and a protoduino process keeps its variables in its instance struct. A module written that way has no `.data` and no `.bss`, so there is nothing in RAM to place or relocate: the kernel hands the thread a pointer, and the module works wherever that points. The rule is enforced at link time rather than left to discipline (6.8).

4.7 Relocation without a relocator

The first module scheme links each module for one slot, which needs no code on the device. To let one file load into any slot, addresses inside the module must be adjusted, and the usual way needs a relocation engine. There is a cheaper way. If slots are aligned to 512 bytes, moving a module changes only the high byte of every byte address and of every word address; low bytes never change and no carry crosses between them. A fix-up then is one position in the stream whose high byte gets a constant added. The host sorts those positions by offset and puts the list ahead of the code, and the loader applies them as the bytes go by, still in one pass. On parts with absolute call instructions, calls from the module into the kernel need no fix-up at all. PROPOSED

4.8 The same core on the host, and the simulator as oracle

The core is plain C with no AVR dependency, so it compiles unchanged for the host. That gives three things: unit tests and fuzzing at desktop speed; a host tool that can say exactly what a device will accept, because it runs the device's code; and, compiled to WebAssembly, the same check inside a Node command-line tool or protosim's web front end. For end-to-end tests, the simulator already contains a trusted ELF reader. Flashing a file through the loader inside simavr and comparing the result with what simavr's own reader produces for the same file tests the loader against an independent implementation. Section 9 reports that comparison.

5 Functional specification

5.1 Use cases

ID	Use case	How it goes	Status
UC1	Flash firmware over a serial line	Reset the board. The loader asks for a file by sending `C`. Send the `.elf` with XMODEM from any terminal program. The loader writes it, stores the record and starts it.	MEASURED
UC2	Recover a board whose application is broken or half-written	The record is missing or damaged, so after reset the loader keeps asking for a file. No programmer is needed on parts with a boot section.	MEASURED
UC3	The application asks for an update	It calls `elfboot_request()`, which sets a flag that survives reset and resets through the watchdog. The loader sees the flag and waits for a file.	PROPOSED
UC4	Check what is installed	Send the same file in verify mode. Bytes are compared with flash instead of written.	PROPOSED
UC5	Ask the device what it is	Send `?` before a transfer. The loader answers with part, page size, window and feature bits.	BUILT
UC6	Install a module while the system runs	The kernel's loader process receives the module file, writes it into a free slot through the service, records it and starts it as a process.	PROPOSED
UC7	List, stop and remove modules	Shell commands on the kernel's loader process. Installed modules start again after a reset.	PROPOSED

5.2 Requirements

ID	The loader ...	Evidence
FR-1	accepts ELF32, little-endian, executable files for the AVR machine type whose program headers start at offset 52, fed in file order. This is what avr-gcc emits.	MEASURED T1, T3, T11, host
FR-2	needs nothing after the last loadable byte. Whatever follows, including XMODEM padding, is ignored.	MEASURED T3, T4

ID	The loader ...	Evidence
FR-3	refuses a file built for another core family from the ELF header alone.	MEASURED T5, T7
FR-4	refuses any flash segment outside the permitted window. The loader's own pages and the image record lie outside the window.	MEASURED host
FR-5	changes nothing in flash until every program header has been accepted.	MEASURED T7
FR-6	invalidates the installed image before changing its first byte, and makes an image valid only after its last byte is written.	MEASURED T1; power cut not simulated
FR-7	has a damaged block sent again, and acknowledges a repeated block without applying it twice.	MEASURED T2
FR-8	optionally writes the EEPROM image contained in the file.	MEASURED T8
FR-9	optionally compares the file's device signature with the chip's. A mismatch prevents the commit. The signature follows the flash data in the file, so by then flash and EEPROM have changed and the device stays in the loader. The host tool makes the same check before sending.	MEASURED T9
FR-10	after a refusal sends a one-byte reason and is ready for a new transfer. If nothing was written, the old image still starts.	MEASURED T5, T6, T7
FR-11	starts an image only if its record is intact, and optionally only if the image's CRC still matches.	record MEASURED; CRC at start BUILT
FR-12	exports a page-write entry to the application. The entry uses the caller's memory only.	MEASURED T10
FR-13	installs a module only into a free slot and only if the module names this firmware build; starts it as a process; starts it again after reset; removes it on request.	link recipe and window MEASURED on host; rest PROPOSED
FR-14	never has to accept a module with static RAM: the module linker script rejects it.	MEASURED

5.3 Start-up behaviour

After reset the loader probes for a sender: it transmits `C`, waits about a second, and repeats. A transfer may begin at any probe. If none begins within the configured number of probes and the image record is intact, the image is started. If the record is not intact, the loader probes for ever. In the prototype the window is always applied and comes to about five seconds with three probes MEASURED. That is too long for a product, so the entry condition is a configuration choice: always wait; wait only after an external reset; or wait only when the application asked or no valid image exists. PROPOSED

5.4 Reasons for refusal

When the loader refuses a file it cancels the transfer and sends one more byte with the reason.

Code	Name	Meaning	Flash changed?
`0x80`	FORMAT	Not a little-endian ELF32 executable for AVR with program headers at offset 52	no
`0x81`	ARCH	Built for another AVR core family	no
`0x82`	LAYOUT	No program headers, or an entry address that does not fit	no
`0x83`	RANGE	A segment falls outside the permitted window, or an offset or size does not fit the build's address width	no
`0x84`	ORDER	Segment data is not in ascending file order, or a block arrived out of sequence	no for the file; possibly for a block
`0x85`	SEGMENTS	More separate segments than the build remembers	no

Code	Name	Meaning	Flash changed?
`0x86`	EMPTY	Nothing to load	no
`0x87`	SINK	A sink refused a byte: today, a device signature that does not match	yes; not committed
`0x88` – `0x8A`	ABI, SLOT, RAM	Module built for another firmware; no free slot; module declares static RAM PROPOSED	no

5.5 Limits

- At most 255 program headers; at most `ELFLD_CONF_MAX_SEGMENTS` separate loadable segments after adjacent ones are merged. Plain firmware needs one.
- File offsets and sizes must fit the build's address width: 64 KB on small parts, 16 MB on large ones.
- Flash segments must arrive in ascending address order, because the page cache only moves forward. avr-gcc output does. A read-modify-write option would lift this. PROPOSED
- The serial path needs 128 bytes of RAM for an XMODEM block and one flash page more.

6 Technical design

6.1 Where the pieces live

```
src/sys/elf/elfld.h  elfld.c  elfld.conf.h     core; also builds on the host
src/sys/elf/module.h  elfmod.c                module descriptor and the kernel's loader process (cut C)
src/cpu/avr/flash.h   flash_spm.c             flash back-end and the client side of the service
boot/elfboot.c  xmodem.h  elfboot.conf.h      the bootloader, with its own build
ld/pd-module.ld                               linker script for modules
tools/pd/                                     Node 22 ESM command-line tool: send, verify, info, abi
```

The Arduino build compiles every source file under `src/`. The bootloader has its own entry code and no C runtime, so it must stay outside that tree.

6.2 The core

```
/* Start a load. Flash bytes outside [lo, hi) are refused before anything is written. */
void    elfld_begin(elfld_t *c, elfld_addr_t lo, elfld_addr_t hi);

/* Push the next n bytes of the file. Returns ELFLD_MORE, ELFLD_DONE or an ELFLD_E_* code;
   after DONE or an error the rest of the file is ignored. */
uint8_t elfld_write(elfld_t *c, const uint8_t *p, uint8_t n);

/* Supplied by the port: store one byte. Return 0 to continue. */
uint8_t elfld_sink(uint8_t space, elfld_addr_t addr, uint8_t byte);
```

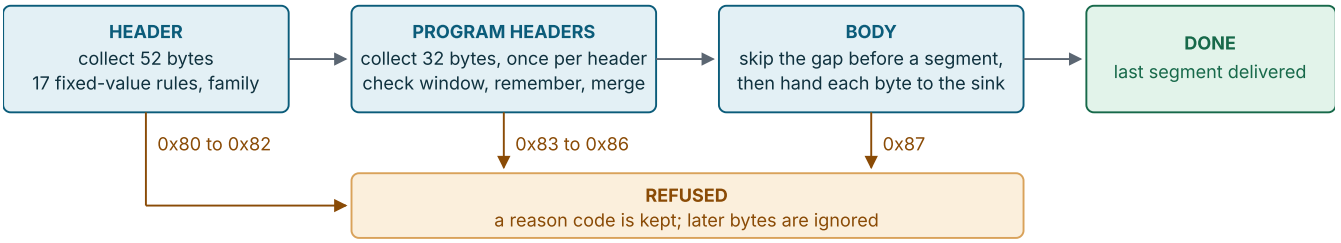


Figure 4. The core's states. Nothing reaches a sink before BODY, which is why a refusal in the first two states changes nothing.

Header rules. Seventeen bytes of the ELF header must each have one exact value: the magic, the class, the byte order, the file type, the machine, the position and size of the program headers. They are checked by a loop over a 34-byte table of offset and value pairs, which is smaller than seventeen comparisons and easy to extend.

What is remembered. The data of a segment arrives after all the headers, so each loadable segment is remembered as three numbers: the count of file bytes to skip before it, the address of its next byte, and the bytes still to deliver. Storing the gap and not the absolute offset removes the need to count file position at all. A segment that continues the previous one, both in the file and in memory, is merged into it. `.text` and `.data` always do, so plain firmware is one remembered segment.

Context layout. The scratch area for header bytes is the first member of the context, so that header bytes and the hot control fields are all within the 63-byte reach of one pointer. On AVR that matters more than it looks: see 6.10.

What building the prototype taught

- **A refused file must not cost the installed image.** The first version erased the record when the first block arrived. A wrong file then left a working board without a valid image. Invalidation now happens with the first page write, after the headers are accepted. Test T7 guards it.
- **Above 64 KB, tables in flash need the far read.** In the ATmega2560's boot section the rule table sits beyond the reach of `LPM`. The loader refused every file until the table was read through `ELPM`.
- **Service code owns no RAM.** When the application calls into the boot section, the loader's variables are the application's memory. The first service copied into the loader's page buffer and would have corrupted the caller. It now works on the caller's buffer alone.
- **Do not pin registers.** An attempt to force the context pointer into the Y register with a local register variable compiled, did nothing on one part and miscompiled on the other. The supported way is the compiler option in 6.10.

6.3 Flash back-end

The sink for flash is a cache of one page. A byte for a different page first commits the cached one: erase, fill the chip's page buffer, write, re-enable the read-while-write section. The cache starts as all `0xFF`, so a partly covered page is padded. The back-end is the only part that differs between chip families:

Back-end	Parts	Notes	Status
SPM from a boot section	megaAVR	Read-while-write must be re-enabled before the application section is read again	MEASURED
SPM from anywhere	classic tinyAVR	The CPU halts during a write; needs the SELFPRGEN fuse; reset vector handling in 6.6	BUILT
NVMCTRL	tinyAVR 0/1/2, AVR Dx	Different mechanism; not designed here	–

The lock bits must leave SPM writes to the application section enabled, or no bootloader of any kind can work.

6.4 Transport: XMODEM-CRC

The bootloader speaks XMODEM with 16-bit CRC. It is decades old and exactly right for this job.

Property	Why it matters here
Built into terminal programs on every platform	The unmodified `.elf` can be sent with no tool of ours installed.
The receiver sets the pace: one block, one acknowledgement	The loader programs pages while handling a block and acknowledges afterwards. No flow-control lines and no receive buffer are needed with a polled serial port.
Blocks are numbered	A block repeated after a lost acknowledgement is recognised and not applied twice.
CRC-16 per block	The same CRC routine also protects the image record and the image, so there is one checksum routine in the loader.

A block is `SOH`, the block number, its complement, 128 data bytes and two CRC bytes. The loader holds a block in RAM until its CRC has passed, and only then feeds it to the core, so damaged bytes never reach flash. One extension is used: after cancelling with two `CAN` bytes the loader sends the reason code. Terminal programs ignore it; the `pd` tool prints it.

The limits are the 128 bytes of RAM a block needs, and that XMODEM cannot tell the sender to stop early. Sending a stripped file, or using the `pd` tool, which stops after the last loadable byte, removes the wasted time. In simulation at 115200 baud a block took 13.3 ms of line time: 0.22 s for example 12 stripped, 3.06 s with its debug information. MEASURED

The runtime loader (cut C) uses the same framing over a protoduino pipe from the serial process. Stop-and-wait is not optional there: programming a page keeps interrupts off for several milliseconds, and a sender that kept transmitting would overrun the receiver.

6.5 Image record and memory map



ATmega328P with the 2 KB boot section, not to scale. Window 30,592 bytes; loader 1,676 to 1,944 bytes used; service stub and magic in the last four bytes.

Figure 5. Flash layout. The record page sits directly below the loader and outside the window.

Bytes	Field	Content
0 – 1	Magic	`'p' 'D'`
2 – 3	Entry	Word address at which the image is started
4 – 6	Image end	End of the written image, rounded up to a page
7 – 8	Image CRC	CRC-16 (XMODEM polynomial) of flash from 0 to the image end
9 – 10	Record CRC	Stored so that the CRC over all eleven bytes is zero

An erased page fails both the magic and the CRC, so “no record” needs no flag. The record after test T1 read `70 44 00 00 80 01 00 13 80 c1 aa`: entry 0, image end `0x000180`, image CRC `0x8013`. That 16-bit value doubles as the identity of the firmware build for modules (6.8). The rest of the record page is reserved for module records.

6.6 Parts without a boot section PROPOSED

On a classic ATtiny, reset always goes to address 0, and nothing protects the loader's pages. The loader sits at the top of flash and makes four provisions.

1. **The reset vector is redirected.** When page 0 of an image passes through the page cache, the loader keeps the image's own start address for the record and replaces the first instruction with a jump to itself. The other vectors stay the image's; the loader uses no interrupts.
2. **Erase runs from the top down, page 0 last.** Before the first page is written, all application pages are erased starting just below the loader. While page 0 still holds the old redirected vector, a reset reaches the loader through it. Once page 0 is erased, everything below the loader is erased, and erased flash executes as a harmless skip instruction until the program counter runs into the loader. This closes the window urboot's documentation warns about.
3. **A landing pad.** Because that skip instruction may skip one word, the loader starts with a one-word no-op.
4. **Its own serial port.** The ATtiny85 has no USART; the build uses a bit-banged port.

The loader's own pages remain unprotected against a runaway application. On these parts the design is an updater, not a guaranteed recovery path, and an ISP programmer remains the last resort. This path compiles and has not executed. The simulator cannot settle point 2: simavr treats `0xFFFF` as an invalid opcode.

6.7 The service entry

```
/* last four bytes of flash:  rjmp bootsvc_write_page ; 'p' ; 'D' */
typedef void (*pd_write_page_fn)(uint32_t addr, const uint8_t *page);

if (pgm_read_byte(FLASHEND - 1) == 'p' && pgm_read_byte(FLASHEND) == 'D')
  ((pd_write_page_fn)((FLASHEND - 3) / 2))(0x6000UL, page);     /* returns when the page is programmed */
```

The entry disables interrupts while it works, because the caller's interrupt handlers live in the section being programmed, and restores the caller's interrupt state on return. It costs 28 bytes in the loader. In simulation an application installed through the loader called it and read its page back correctly MEASURED. Two limits apply to that result: simavr does not enforce the rule that SPM must run from the boot section, so the test shows the calling convention and not the hardware permission; and on parts above 128 KB the entry lies beyond a 16-bit function pointer and needs a small far-call trampoline, which is not built.

The planned form is a short table, not a single entry: write a page, begin a load, feed a load, report information. With the core exported this way, the kernel's module loader carries no second copy of it. PROPOSED

6.8 Modules PROPOSED

A module is an executable linked into a flash slot. Its first bytes are a descriptor, and the ELF entry point is the descriptor's address, so the standard header field tells the loader what was installed.

```
#include <protoduino.h>
#include <sys/process.h>
#include <sys/elf/module.h>

PROCESS_DEFINE(blink, "blink", 2,
  uint8_t  pin;                 /* all state lives in the process struct */
  uint16_t ticks;
);
PD_MODULE(blink);               /* descriptor: magic, ABI, firmware id, &process_type_blink */

PROCESS_THREAD(blink, ev, data)
{
  blink_t *self = PROCESS_SELF(blink);
  PROCESS_BEGIN();
  /* ... ordinary protoduino process code, calling kernel functions directly ... */
  PROCESS_END();
}
```

```
# the kernel's functions, as addresses the module may link against
avr-nm firmware.elf | pd abi > kernel-abi.ld          # PROVIDE(process_post = 0x00001a2c); ...

avr-gcc -Os -mmcu=atmega328p -nostartfiles -nostdlib \
        -Wl,-T,pd-module.ld -Wl,-T,kernel-abi.ld \
        -Wl,-e,pd_module -Wl,--defsym=__pd_kernel_id=0x8013 \
        -o blink.elf blink.c
```

A reduced form of this recipe was linked and loaded on the host MEASURED. The result is an 848-byte file with one flash segment at the slot address, its entry point on the descriptor, and calls into the kernel as plain absolute `call` instructions. The core accepted it when the window was the slot it was linked for, and refused it with `RANGE` for a different slot; the firmware image offered to a module slot was refused the same way. The module linker script turns static variables into a link error: *module has static variables: keep state in the process struct*.

Binding to the kernel

Scheme	How a module finds kernel functions	Cost on the device	Trade-off
Exact first version	Linked against the addresses of one firmware build. The descriptor carries that build's identity: the image CRC from the record. The loader compares the two and refuses on a mismatch.	None. No symbol table, no names, no table of entries.	Modules must be re-linked after any firmware change. A stale module is inert, not a crash.
Table option	The kernel exposes an append-only table of jumps at a fixed address; modules link against the table slots.	Four bytes per exported function, and one extra jump per call.	Modules survive firmware updates while the table only grows.

Slots, records and life cycle

- **Slots.** Pages between the end of the firmware image and the record page. The information reply names the next free slot, so the host can link for it. With 4.7, any slot.
- **Records.** A few bytes per module in the record page: state, first page, page count, descriptor address, CRC.
- **Install.** The descriptor is the first thing in the first segment, so the loader can check its magic and firmware id while the first page is still in RAM, before any write. Pages go through the service. The record is written last. Then `process_new()` with memory from a module pool, and `process_start()`.
- **Start-up.** For every intact record whose firmware id matches, start the process again.
- **Remove.** `process_destroy()`, which delivers the exit event, then clear the record.

Rules and hazards

- No static variables. No interrupt handlers: the vector table belongs to the kernel.
- A process struct is at most 255 bytes, because `process_type_t.size` is one byte.
- A callback a module registers with the kernel, such as a UART receive handler, points into the module. It must be removed in the module's exit handler. The kernel cannot check this.
- While a page is programmed, interrupts are off for milliseconds. Installing a module disturbs time-critical processes.
- A signature over the module, checked before the record is written, would fit in the descriptor. It is reserved, not designed.

6.9 Compile-time configuration

Configuration follows the pattern protoduino already uses in its `*.conf.h` files: every setting has a default and can be overridden. Three rules apply. Derive from the part wherever possible: address width from `FLASHEND`, expected core family from `__AVR_ARCH__`, page size from `SPM_PAGESIZE`. A feature that is off costs nothing. And fail the build, not the device: a missing setting is an `#error`, and a loader that outgrows its boot section must fail at link time.

Setting	Default	Effect	Flash, 328P
`ELFLD_CONF_ADDR_BITS`	from the part	16 or 24 bits for offsets and addresses	see 6.10
`ELFLD_CONF_MAX_SEGMENTS`	3	Separate segments remembered; 6 or 9 bytes of RAM each	–
`ELFLD_FEAT_ARCH_CHECK`	on	Refuse files for another core family	12 B
`ELFLD_FEAT_COALESCE`	on	Merge adjacent segments	62 B
`ELFLD_FEAT_EEPROM`	off	Write the EEPROM image	+80 B
`ELFLD_FEAT_SIGNATURE`	off	Compare the device signature	+74 B
`ELFLD_FEAT_RAM_EXTENT`	off	Record the RAM the file claims; for modules	host only
`ELFLD_FEAT_WINDOW_LO`	off	A lower bound on the window; for module slots	host only
`ELFBOOT_FEAT_IMAGE_CRC`	on	Store the image CRC in the record	26 B
`ELFBOOT_FEAT_VERIFY_BOOT`	off	Re-check the image CRC at every start	+46 B
`ELFBOOT_FEAT_INFO`	off	Answer `?` with a description line	+68 B
`ELFBOOT_FEAT_SERVICE`	off	Export the page-write entry	+28 B
`ELFBOOT_VIRTUAL_RESET`	off	Part has no boot section (6.6)	tiny only
`ELFBOOT_SOFT_UART`	off	Bit-banged serial for parts without a USART	tiny only
`ELFBOOT_CONF_WINDOW`, `BAUD`	3, 115200	Probes before starting the image; line speed	–

Each figure is the difference from the default build of 1,676 bytes when that one setting is flipped MEASURED. With every optional feature off the loader is 1,576 bytes; with everything on, 1,944. Named profiles on top of these settings, for a minimal bootloader, a standard one, a full one and the kernel-side loader, would give users one line to choose. PROPOSED

6.10 Measured budgets

Build	Flash	RAM
ATmega328P, optional features off	1,576 B	343 B
ATmega328P, default	1,676 B	349 B
ATmega328P, everything on	1,944 B	359 B
ATmega1284P, default	2,164 B	491 B
ATmega2560, default	2,168 B	491 B
ATmega2560, everything on	2,488 B	504 B
ATtiny85, bit-banged serial BUILT	1,736 B	285 B

Where the 1,676 bytes go	Flash
XMODEM receiver, session, record write	506 B
Core: delivery loop and flash sink	348 B
Core: program headers	242 B
Core: header check, rule table, begin	190 B
Page programming and commit	184 B
CRC-16, for blocks and for flash	98 B
Serial get and put; start-up; image start	108 B

What fits where. The ATmega328P offers boot sections of 0.5, 1, 2 and 4 KB. Every 328P build fits the 2 KB section, with 104 bytes to spare when everything is on. None fits 1 KB: the default is about 650 bytes over. The ATmega2560 offers 1, 2, 4 and 8 KB; the default build is 120 bytes over the 2 KB section and so needs 4 KB today. RAM on the 328P is the XMODEM block (128), the page buffer (128), the core context (84) and nine bytes of flags.

Why it is this big, and what would shrink it. The code is ordinary C and the cost is spread evenly: ten to thirty bytes per source line, mostly 16-bit arithmetic on fields reached through a pointer. Nothing has been tried from this list, in order of expected yield:

1. Write the per-byte path and the SPM sequences in assembly, as Optiboot and urboot do.
2. Replace the 128-byte block with a smaller frame, which also removes the buffer-then-apply loop and suits the small ATtinys.
3. Build with a newer avr-gcc. The repository's `platformio.ini` already points at one; only 7.3.0 was available here.
4. Take the minimum cut: without merging, the family check and the image CRC the loader is 100 bytes smaller.

One compiler option already mattered. By default avr-gcc kept the context pointer in the X register, which has no displacement addressing, and spent three instructions on each field access. With `-mstrict-X` the header check alone fell from 154 to 120 bytes. MEASURED

Time. Line time dominates: 13.3 ms per block at 115200 baud in simulation, about 9.6 KB per second. Real hardware adds the time to erase and program each page, a few milliseconds each, which simavr does not model.

7 Fixes for current protoduino caveats

These came out of reviewing the UART and serial layers as a base for the loader. Each was checked by running protoduino's real sources under protosim, except where marked as read. None has been applied. Four are defects; the rest are small.

#	Where	What happens	Evidence
1	`src/cpu/avr/uart_private_impl.h:69`	`uartN_baudrate()` reports twice the real rate whenever the double-speed bit is off. The line rate itself is correct.	**MEASURED** request 57600: registers give 58,823, function returns 117,647. Request 300: real 300, reported 600.
2	`uart_private_impl.h:81–87`	`uartN_close()` ends with an unconditional `sei()`. Called with interrupts disabled, it switches them on under the caller.	**MEASURED** interrupt flag 0 before the call, 1 after.
3	`src/sys/serial/serial_private_impl.h:293`	`serialN_flush()` tests the interrupt flag the wrong way round compared with its own documentation. With interrupts off it enables the transmit interrupt, which cannot fire, and sends nothing.	**MEASURED** ten bytes queued with interrupts off: after 60,000 calls nine are still unsent. With interrupts on all ten go out.
4	`docs/loader.md`	Describes a loader that cannot work on AVR (module code in RAM, `-fpic`) and an API that is not in `src/`. The README links to it as experimental.	**READ** avr-gcc prints "-fpic is not supported". No loader source exists in the tree.
5	`uart_private_impl.h:105–109`	The frame-error branch in `uartN_rx_error()` can never run: the flag is masked out three lines earlier. Skipping frame errors is a documented choice; the dead branch is the only defect.	**READ**
6	UART header documentation	Nothing says that the polled receive calls and the receive interrupt must not be used together. With both active the stream splits between them.	**MEASURED** input `hello`: the callback received `hlo`, the poller `el`.
7	`uart_private_impl.h`	The interrupt handlers and callback pointers are always compiled in. A bootloader that wants the same register mapping without vectors cannot have it.	**READ** The polled calls themselves work with interrupts off **MEASURED**: input `hello` echoed in full.
8	`platformio.ini`	`lib_deps` is an absolute path on one Windows machine, so the project builds nowhere else as checked in.	**READ**
9	`uart_private_impl.h:44, :162`	A baud rate of 0 divides by zero in `openex()`. A `#pragma message` prints on every build of every UART instance.	**READ**

Proposed changes

```
/* 1. uart_private_impl.h, openex(): remember which divisor is in force */
  uint8_t  samples = 8;                                  /* double speed: clock / 8 */
  uint16_t baud_setting = (F_CPU / 4 / baud - 1) / 2;
  __UCSRA__ = 1 << __U2X__;
  if (((F_CPU == 16000000UL) && (baud == 57600)) || (baud_setting > 4095)) {
    baud_setting = (F_CPU / 8 / baud - 1) / 2;
    __UCSRA__ = 0;
    samples = 16;                                        /* normal speed: clock / 16 */
  }
  ...
  CC_TMPL_VAR(baudrate) = F_CPU / ((uint32_t)samples * (baud_setting + 1));

/* 2. uart_private_impl.h, close(): leave the caller's interrupt state alone */
void CC_TMPL_FN(close)(void)
{
  CC_ATOMIC_RESTORE()
  {
    CC_TMPL_VAR(baudrate) = 0;
    __UCSRB__ &= ~(_BV(__RXEN__) | _BV(__TXEN__) | _BV(__RXCIE__) | _BV(__UDRIE__));
  }
}

/* 3. serial_private_impl.h, flush(): poll the port when interrupts are OFF */
  if (bit_is_clear(SREG, SREG_I))
  {
    if (!CC_TMPL2_FN(tx_is_ready)())
      return cnt;
    CC_TMPL2_FN(tx_write8)(ringb8_get(&VAR_TX));
    return cnt - 1;
  }
  CC_TMPL2_FN(tx_enable_int)();
  return cnt;
```

- **4.** Replace `docs/loader.md` with a short page that points here, and mark the README entry as a design, not a feature.
- **5.** Delete the dead branch. If frame errors are wanted later, report them from the receive interrupt, where the flags are read before the data register and are valid; put it behind `UART_CONF_REPORT_FRAME_ERRORS`, default off, so behaviour stays as it is.
- **6.** One paragraph in `uart_private_header.h`: use the callbacks with interrupts on, or the polled calls with interrupts off, never both.
- **7.** Add `UART_CONF_ISR`, default 1. With 0, the handlers, the callback setters and the receive-interrupt enable in `openex()` are left out, and the bootloader can use the same template.
- **8.** A relative library reference in `platformio.ini`.
- **9.** Return `ERR_SERIAL_BAUD` for a zero rate; put the message behind a verbose-build switch.

Findings from my first review that testing withdrew

They are listed so the record is straight. The polled receive calls do work after `open()` when interrupts are off. `rx_clear_errors()` does clear an overrun under simavr; on a real chip those flags are read-only and clear when the data register is read, so the call is harmless there. The transmit path through the data-register-empty interrupt is sound: 18 bytes went out in 19 handler calls. The 57600 special case is the deliberate Arduino-compatible behaviour. The read-modify-write on the control register can at worst cause one extra interrupt that switches itself off.

8 What protosim needs

protosim already loads a bootloader beside an application and can drive one over the simulated UART. Testing this loader with it, not with a separate harness, needs the following. All are proposals.

#	Change	Why	Evidence
P1	Binary input for the UART: `--uart0-in-file`, or an XMODEM sender beside the STK500 one	`--uart0-in` takes a C string. An ELF file contains zero bytes and cannot be passed.	READ
P2	Pace injected bytes from the UART's configured rate	Bytes are injected every millisecond whatever the baud rate. At 9600 a byte takes 1.04 ms, so bytes queue inside simavr and the receive flag stays set with no interrupt pending.	MEASURED the flag stayed visible for 568 consecutive polls at 9600; never at 19200.
P3	Set simavr's `reset_pc` when `--bootloader` is given	Only the initial program counter is set today. A watchdog or software reset would restart at address 0, so the "application asks for an update" path cannot be tested.	READ protosim never assigns `reset_pc`. The harness set it and reset correctly.
P4	`--expect-flash <file.elf>`: compare flash with simavr's own load of a file at exit	This is the oracle test of section 9, as a one-line check for CI.	MEASURED in the harness
P5	Page size from the part in the STK500 uploader	The uploader assumes 128-byte pages, so the automatic upload suits ATmega328P-class parts only.	READ
P6	Work around simavr's ELF reader at the end of flash	It refuses a chunk that ends on the last flash byte ("would overflow buffer"), so a service stub in the last four bytes is silently not loaded.	MEASURED the service test had to place its stub four bytes lower.

9 Verification

9.1 End to end in simulation

The loader ran inside simavr. A harness sent each file with XMODEM as a terminal program would, at 115200 baud, and afterwards compared the simulated flash below the record page with what simavr's own ELF reader produces for the same file.

Test	Scenario	Outcome
T1	328P: unmodified avr-gcc ELF, 12,860 bytes, clean line	Flash identical to the reference; record written; application reached
T2	328P: block 3 damaged in transit, acknowledgement of block 5 lost	One block sent again after a negative acknowledgement, one duplicate dropped; flash identical
T3	328P: protoduino example 12, 29,404 bytes with debug information	Flash identical; 3.06 s of line time
T4	328P: the same example stripped, 2,048 bytes	Flash identical; 0.22 s
T5	328P, empty chip: an ATmega2560 file	Refused `0x81`; flash untouched
T6	328P, empty chip: random bytes	Refused `0x80`; flash untouched
T7	328P: install, reset, then offer an ATmega2560 file	Refused `0x81`; the installed application intact and started again after the window

Test	Scenario	Outcome
T8	328P, all options: file with EEPROM data and a matching signature	Flash identical; EEPROM holds `01 02 03 04 05`
T9	328P, all options: an ATmega168P file, same core family, other device	Refused `0x87` at the signature; no record written; the loader keeps control
T10	328P: an installed application programs a page through the service entry	Page reads back as written; the caller's interrupt state restored
T11	2560: small image	Flash identical; record written; application reached
T12	2560: 106,830-byte image reaching far flash, 839 blocks	Flash identical; 11.9 s of line time

9.2 On the host

The same core, compiled natively, was fed real files in chunks of random size and its output compared with `avr-objcopy`.

- Flash and EEPROM images identical for the test program built for ATmega328P, ATmega2560 and ATtiny85; identical for example 12, for its stripped form, for one-byte-at-a-time feeding, and for the 106 KB image.
- The signature bytes extracted from the three files were those of the three parts.
- Refused as designed: a file for another family (`0x81`), an image larger than the window (`0x83`), a non-ELF file, a relocatable `.o` and an x86-64 object (all `0x80`). A truncated file leaves the core waiting for more.
- A module was accepted in its own slot and refused in another; the firmware was refused in a module slot.

9.3 Not verified

- Anything on hardware, including fuse and lock-bit settings and real programming times.
- Loss of power during an update. The transaction argument in 4.4 is by construction.
- The ATtiny path: it has never executed.
- The image CRC check at start-up, the information reply, and the service entry on the ATmega2560.
- The module layer beyond linking and the window check.

9.4 Test plan from here

- **Host unit tests in CI** for the core, with the files above as fixtures, and a fuzzer on `elfld_write()`. The core is small, pure C and takes untrusted input, which is the ideal fuzzing target.
- **protosim end to end** per supported part, once P1 to P4 exist, with the oracle comparison as the pass criterion.
- **Hardware**: an Uno and a Mega 2560, with an ISP programmer at hand for fuses. A relay on the supply to cut power at random points of an update, hundreds of times, checking that the board always answers afterwards.

10 Roadmap, risks and open questions

Step	Deliverable	Done when
M0	The fixes of section 7	The probes in `prototype/probes` give the corrected results
M1	Core in `src/sys/elf`, host tests, fuzzing	Fixtures pass in CI on the host and for two AVR parts
M2	Bootloader for ATmega328P and ATmega2560; protosim P1 to P4	T1 to T12 run in protosim; first boards flashed with a terminal program

Step	Deliverable	Done when
M3	Service table, `elfboot_request()`, information reply; the `pd` tool; PlatformIO upload through it	`pio run -t upload` sends the `.elf`; the power-cut rig passes
M4	Modules, exact binding; the kernel's loader process	A module is installed over serial, runs as a process, survives reset and can be removed
M5	ATtiny path on hardware; relocation by fix-up stream; signed modules; a size pass in assembly	Each on its own evidence

Risks

- **Size.** The 1 KB boot section may stay out of reach. The design does not depend on it; a 2 KB section costs the application 1 KB.
- **ATtiny recovery.** Without hardware protection for the loader's pages, a faulty application can destroy it. The claim for these parts is deliberately weaker.
- **Exact binding.** Convenient while developing, awkward for distributing modules. The table scheme is the answer and costs flash.
- **Interrupt blackout.** Flash writes at run time stall the system for milliseconds at a time.
- **Old compiler.** All sizes are from avr-gcc 7.3.0. A newer one may be better or worse.

Open questions

- Which small part is meant by “ATtiny8”? The answer decides whether the small-frame transport is needed early.
- Should updates be authenticated, and if so with what? A keyed hash costs several hundred bytes.
- Which binding scheme is the default for modules: exact or table?
- Should the loader wait after every reset, or only on request? It changes start-up time by seconds.
- Where should the bootloader's build live, given that it cannot be under `src/`?

A ELF fields the loader reads

Where	Offset	Field	Use
ELF header	0 – 3	magic	Must be `7F 45 4C 46`
	4, 5	class, data	Must be 1 and 1: 32-bit, little-endian
	16	`e_type`	Must be 2, an executable. A relocatable `.o` is refused here.
	18	`e_machine`	Must be 83, AVR
	24	`e_entry`	Start address; for a module, the address of its descriptor
	28	`e_phoff`	Must be 52: program headers directly after the ELF header
	36	`e_flags`	Low 7 bits name the core family; compared with `__AVR_ARCH__`
	42, 44	`e_phentsize`, `e_phnum`	Must be 32; count of program headers, below 256
Program header	0	`p_type`	Only 1, a loadable segment, is considered
	4	`p_offset`	Where the segment's bytes are in the file; must not go backwards
	8, 20	`p_vaddr`, `p_memsz`	Run address and size in memory; used only to record a RAM extent
	12	`p_paddr`	Load address. Its top byte selects the memory; the rest is the address there.
	16	`p_filesz`	Bytes to deliver; a segment with none is skipped

For the ATmega328P test program avr-gcc wrote six loadable segments: code at load address 0; initial `.data` at run address `0x800100` but load address `0x136`, directly after the code; `.bss` with no file bytes; EEPROM at `0x810000`; fuses at `0x820000`; signature at `0x840000`. **MEASURED**

B Reproducing the measurements

Everything is in `docs/elfloader/prototype/`. It needs `gcc-avr`, `avr-libc`, `libelf-dev` and protosim built with its simavr submodule.

```
# core on the host, against a file built with avr-gcc
gcc -std=c99 -O2 -DELFLD_FEAT_EEPROM=1 -DELFLD_FEAT_SIGNATURE=1 -o host_test host_test.c elfld.c
./host_test app.elf 5 out.bin
avr-objcopy -O binary -j .text -j .data app.elf ref.bin && cmp out.bin ref.bin

# the bootloader, and its size
./build.sh m328 atmega328p 0x7800
./build.sh m2560 atmega2560 0x3F000 -DELFBOOT_FEAT_SERVICE=1

# end to end in simavr
S=../../../../protosim/libraries/simavr/simavr
gcc -O2 -o sim_harness sim_harness.c -I$S/sim -I$S/sim/avr \
    $S/obj-x86_64-linux-gnu/libsimavr.a -lelf -lm -lpthread
./sim_harness atmega328p out/m328.elf app.elf --corrupt 3 --drop-ack 5
```

Recorded outputs are in `results-sizes.txt` and `results-tests.txt`. The UART and serial findings of section 7 come from `probes/uart_probe.c` and `probes/serial_flush_probe.c`, built against protoduino's own sources and run with `protosim --uart0-in … --uart0-out …`.

Sources

- Contiki, `core/loader/elfloader.c`, `symtab.c`, `cle.c`: github.com/contiki-os/contiki. Removal: github.com/contiki-ng/contiki-ng, commit 6220aea14.
- simple-avr-bootloader: github.com/m3y54m/simple-avr-bootloader
- Optiboot: github.com/Optiboot/optiboot; version 8 notes with `do_spm()`: forum.arduino.cc/t/optiboot-now-at-version-8-0/547763
- urboot, on erased flash and the first-page hazard: github.com/stefanrueger/urboot. Optiboot on ATtiny: github.com/SpenceKonde/ATTinyCore, `avr/extras/Ref_Optiboot.md`
- micronucleus and its reset-vector patch: github.com/micronucleus/micronucleus
- Zephyr LLEXT and symbol link identifiers: docs.zephyrproject.org/3.7.0/services/llext/api.html
- Microchip, ATmega48A/88A/168A/328P family, “Self-Programming the Flash” and “USART Control and Status Register 0 A”: onlinedocs.microchip.com
- avr-libc memory sections: the avr-libc user manual, “Memory Sections”
- protoduino and protosim sources at the branch named on the cover; simavr at commit a6fc4cc.